

Call for Expressions of Interest	
<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Learning algorithm for a soft robotic wearable device for rehabilitation
<i>Company:</i>	N/A
Contact Details	
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Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☒ If Applicable: The Soft robotics wearable device is ready to use. It is currently in the Mixed Reality Lab.
<i>Relevant research areas (if applicable)¹:</i>	Machine learning, robotics, soft robotics
Project Background Information	
A quarter of adults over the age of 25 will suffer a stroke. Debilitating upper limb impairment occurs in 50-80% of acute cases and remains in 40-50% of chronic cases. Lack of access to rehabilitation remains a significant challenge for stroke survivors, hindering their recovery and quality of life. Wearable soft robotic devices are a promising solution to bridge this gap, offering potential advantages such as affordability and adaptability. However, effectively controlling these devices during dynamic interactions with the user's body remains a challenge. Automating physiotherapy sessions using a wearable robot could significantly reduce the physical strain on healthcare professionals and provide a low-cost and intrinsically safe robotic device for patients.	

¹ An overview of different research areas in the school of computer science can be found [here](#).



Figure 1 Soft robotic wearable device

Our pneumatic wearable soft robotic rehabilitation [device](#) for the upper limb (see Fig.1) aims to tackle that need. Due to the inherent adaptability of soft robotics and the proposed inertial measurement unit (IMU) within a microcontroller-based control system the new robotics system can offer personalised, adaptive rehabilitation exercises, paving the way for transforming stroke rehabilitation practices. The device assists the user in elbow flexion, delivering an 80% efficiency torque transmission ([see relevant paper](#)). Pressurised air is sent to the soft actuator (in yellow in Fig.1) which expands its pockets causing bending of the actuator itself and of user arm attached to the device. In the participant experiments, safety is ensured by keeping the internal pressure of the device low and thus limiting the forces the device imparts on the user's arm. However, this approach hinders functionality as some users need higher forces for arm bending. Hence there is a need for safe and effective control schemes; this will be the goal of this project.

The project will deliver a machine learning algorithm to adapt the bending action to different users. The user data needed for the training of the algorithm has already been collected by a PhD student working on the device. Traditional control approaches based on mathematical models can fall short of accurately describing the bending behaviour of soft robotics with their strong nonlinearities. Learning approaches are especially important given the big variability (the mean force and torques delivered during participant tests varied by $\pm 50\%$) introduced by different body types that change how torque is transferred. The algorithm will be implemented on the ESP32 microcontroller integrated in the device and it will use the output of the IMU and air pressure sensors. The algorithm will be tested in the wearable soft robot currently in the mixed reality lab.

Requirements

The group must provide:

- A working implementation of algorithm as detailed above using the pre-collected user data for training
- To support the claim that it works for multiple users a real-life demonstration is needed
- Algorithm design and testing documentation must also be included

Skills	
<i>Highly desirable:</i> Excellent programming skills. Strong work ethic and the ability to work well in a team. Creativity, curiosity and interest in robotics.	<i>Desirable:</i> Knowledge of machine learning or interest in it.
Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.